



Probability Tutorial

Probability is one of the most widely used mathematical tools, employed not only in characterizing the statistics of experimental results, but in economic predictions, weather forecasts, sports analyses, medical diagnoses, and political prognostication. The basic idea of assigning an event a number from 0 to 1 (or 0% to 100%), with 0 representing no chance of an event's occurrence and 1 representing its absolute certainty, is a numerical realization of basic intuitions about outcomes being likely or unlikely, decisions being risky or safe, actions being bold or cautious.

However, as one of the most widely used mathematical tools, probability is also one of the most widely misused. Failures in understanding the rules and assumptions underlying the assignment and manipulation of probabilities have led to unjustified scientific claims, loss of life, and financial ruin. That probability appears to dovetail so neatly with human intuition belies the fact that people's intuitions are easily misled into assigning probabilities that do not reflect frequencies of real events.

Conditions for Assigning Probabilities

Probability is properly employed when one can enumerate a set of all possible mutually exclusive outcomes of some event. A coin is flipped; the outcome is heads or tails. The weather at noon tomorrow on campus is sunny or rainy or neither sunny nor rainy. A six-sided die is rolled, and the result is 1, 3, 5, or an even number. The "mutually exclusive" condition means that a result can fall into *only one* of the enumerated categories – one cannot roll a die and get both a 3 and an even number. The "all possible" condition means that a result *must* fall into one of the enumerated categories – there are no meteorological conditions that are not sunny, rainy, or neither sunny nor rainy. Note that there are always many sets of possibilities one can choose to categorize an event; one could classify the outcomes of a die roll as a prime number or not a prime number, the weather could be classified as snowing or not snowing.

Once an appropriate set has been enumerated, each outcome, say A, B, or C, can be assigned a probability $p(A)$, $p(B)$, $p(C)$, where each probability, p , is a number from 0 to 1. Given that the enumeration has listed all possible mutually exclusive outcomes, the sum of the probabilities equals 1: $p(A) + p(B) + p(C) = 1$. This equality is one of the fundamental axioms of probability:

The sum of the individual probabilities for a set of all possible mutually exclusive outcomes of an event is 1.

While this mathematical axiom will prove useful, the individual probabilities, such as $p(A)$, $p(B)$, and $p(C)$, can only be specified by appealing to physical considerations.

Assigning Probabilities to Events

The consideration that has been used most successfully in assigning probabilities to events is the identification of a physical symmetry. The bilateral symmetry of a coin means that it has no preference for landing on one side or the other. Similar considerations apply to the rotational symmetry of a die*, and to the fact that facedown playing cards all look the same. The advantage of identifying a physical symmetry is that one can use it to enumerate a set of mutually exclusive outcomes for an event that should each have the same probability of occurring; they are “equiprobable”. A rolled six-sided die should have the same probability, p , for each of its sides rotating to the top. Since $p(1) + p(2) + p(3) + p(4) + p(5) + p(6) = 6p = 1$, we can calculate that $p = 1/6$. Finding a set of physically equiprobable outcomes, and using the fact that they all sum to 1, is a reliable way to figure out the probability of the individual outcomes.

Of course it is rare that we are fortunate enough to be able to identify a physical symmetry for classifying events. More often the physical considerations that enter into the probability of an event are too complicated to be able to identify a set of equiprobable outcomes. In these cases, the most common method for assigning probabilities is to look at the frequency of events. What is the chance of an earthquake of 5.0 or greater on the Richter scale occurring in New York City next year? Historically, only 2 such earthquakes have occurred in the 300 years or so of seismic record keeping. So the probability should be around $1/150$. A common goal of scientific experiments is determining how often some event occurs. We could have determined that there is a $1/6$ chance of a die rolling a 3 by repeatedly rolling a die and seeing how frequently 3 turns up. This process is a manifestation of scientific induction.

It is critical to realize that the method of assigning probabilities based on observed occurrence rates rests on the assumption of a temporal symmetry, namely that the past conditions that produced events with certain frequencies persist in the present and future. Given that seismic conditions change very slowly, using the frequency of earthquakes in the area for the past few hundred years to infer the probabilities of earthquakes today is a reasonable assumption with which to begin. However, physical conditions change; in the geological past, specific faults would have been considerably more active. An exceptionally common mistake in assigning probabilities is to use the frequency of past events as a guide for present probabilities when the conditions that produced the past frequency have changed.[†]

* Symmetries can be misleading: a loaded die is one with external rotational symmetry but an internally asymmetric density that causes certain sides to rotate to the top more frequently.

[†] In the middle of the last decade, large sums of money were lent to many people for the purpose of purchasing homes in the United States. Much of the money was lent under the assumption that no more than 10% of the loans would fail to be repaid, because historically, the default rate for U.S. housing loans had never exceeded 10%. Many lenders failed to realize that the criteria for qualifying for these loans had been relaxed to an unprecedented degree, and in some cases the default rate exceeded 50%. The false assumption of temporal symmetry with regard to the frequency of housing loan defaults was a central precipitator of a global economic crisis.

Information and Conditional Probability

Learning new information often allows us to change assigned probabilities. The probability that a particular person will be struck by lightning is low, but the probability we assign to that event will increase if we get the additional information that the person is currently on the summit of a mountain and there is a lightning storm going on.

The probability of A given that B is true is called the conditional probability of A given B, written $p(A|B)$. In the previous example, we saw that the probability of being struck by lightning, or $p(\text{struck by lightning})$, is less than the conditional probability $p(\text{struck by lightning} | \text{on a mountain in a lightning storm})$.

Calculating Probabilities of Multiple Events

Once one has assigned probabilities to individual events, A and B, it is possible to calculate the probability of combinations of these events, in particular the probability that *both* A and B happen, $p(A \& B)$, and the probability that *either* A or B (or both) happen, $p(A \text{ or } B)$. To do this we must know the conditional probability $p(A|B)$. We will not deal here with the general case, but rather two cases of conditional probability that one frequently encounters in probability calculations: $p(A|B) = 0$ and $p(A|B) = p(A)$.

The case $p(A|B) = 0$ is another way of stating that A and B are mutually exclusive events; if B occurs then the probability of A occurring is 0. Usually it is straightforward to identify situations in which A and B do not occur together. Mutual exclusivity means that the probability of both occurring is 0 and the probability of either occurring is the sum of the individual probabilities. Summarizing:

If A and B are mutually exclusive, i.e. $p(A|B) = 0$, then:

$$p(A \text{ or } B) = p(A) + p(B) \text{ and}$$

$$p(A \& B) = 0$$

These formulas can be extended to cases of three or more mutually exclusive events.

The case $p(A|B) = p(A)$ is another way of stating that the probability of A and B are independent; whether B happens or not, the probability of A is unchanged. Unlike the mutually exclusive case, it is often very difficult to know when probabilities are independent. For example, if the stock market went up yesterday, does that make it any more or less likely it will go up today? If a basketball player makes a shot is it more or less likely he makes his next shot? If it rained today does that make it any more or less likely that it will rain tomorrow? The belief that independent events are not independent is a common misconception known as the gambler's fallacy. A string of favorable or unfavorable results in independent rounds of a game of chance is woven into an erroneous narrative where the events are interconnected through a spell of good or bad luck. The opposite fallacy, believing events are independent when they are not, can be equally ruinous.[‡]

[‡] Suppose that three pressure relief valves, individually able to prevent an explosion at a power plant, are judged to have a 1% chance of failing in an emergency. If the probability of failure for each valve is thought to be

The probability of two independent events both occurring, such as the first roll of a die turning up a 3, $p(A)$, and the next roll turning up a 5, $p(B)$, is the product of the two individual probabilities. To see this, consider that there are 6 equiprobable outcomes of the first roll and 6 equiprobable outcomes of the second roll, for a total of 36 equiprobable two-roll combinations, only one of which is a 3 followed by a 5; $p(A \& B) = p(A)p(B)$ or $1/36 = (1/6)(1/6)$.

		Second Roll:					
		1	2	3	4	5	6
First Roll:	1	1-1	1-2	1-3	1-4	1-5	1-6
	2	2-1	2-2	2-3	2-4	2-5	2-6
	3	3-1	3-2	3-3	3-4	3-5	3-6
	4	4-1	4-2	4-3	4-4	4-5	4-6
	5	5-1	5-2	5-3	5-4	5-5	5-6
	6	6-1	6-2	6-3	6-4	6-5	6-6

While it is possible to list all possible equiprobable two-dice rolls, as the number of independent events increases, it is not pragmatic. But just as the independent probabilities for each event multiply, so does the number of equiprobable outcomes, e.g., for ten rolls there are 6^{10} equiprobable outcomes and the probability of rolling, say, 3, 5, 3, 5, 3, 5, 3, 5, 3, 5 is $1/6^{10}$.

To determine the probability of *either* of two independent events occurring, we use the fact that $(A \text{ or } B)$ and $((\text{not } A) \& (\text{not } B))$ form a complete set of mutually exclusive outcomes, i.e. either A or B happens or neither of them happens, so that $p(A \text{ or } B) + p((\text{not } A) \& (\text{not } B)) = 1$ or

$$p(A \text{ or } B) = 1 - p((\text{not } A) \& (\text{not } B)). \quad (\text{Equation 1})$$

Since $(\text{not } A)$ and $(\text{not } B)$ are, like A and B, independent events we also know

$$p((\text{not } A) \& (\text{not } B)) = p(\text{not } A)p(\text{not } B). \quad (\text{Equation 2})$$

Finally, since A and $(\text{not } A)$ are a complete set of mutually exclusive outcomes (as are B and $(\text{not } B)$), we have

$$p(A) + p(\text{not } A) = 1 \quad \text{and} \quad p(B) + p(\text{not } B) = 1. \quad (\text{Equation 3})$$

Combining Equations 1, 2, and 3, we find:

$$p(A \text{ or } B) = 1 - p((\text{not } A) \& (\text{not } B)) = 1 - p(\text{not } A)p(\text{not } B) = 1 - (1 - p(A))(1 - p(B))$$

This relation is somewhat complicated, but is frequently invoked in probability calculations.

independent, one would calculate that there is a $(0.01)(0.01)(0.01) = \text{one-in-a-million}$ chance that none of them work in an emergency. In reality it is likely that the conditions that cause one valve to fail, say a humid environment that rusts a valve shut, cause all three valves to fail, meaning the failure probabilities are not independent and the possibility of an explosion in an emergency is closer to 1% than one-in-a-million.

Summarizing the rules for independent events:

If A and B are independent, i.e. $p(A|B) = p(A)$, then:

$$p(A \& B) = p(A)p(B) \text{ and}$$

$$p(A \text{ or } B) = 1 - (1 - p(A))(1 - p(B))$$

Again these formulas can be extended to cases of three or more independent events.

The Probability of Rare Events

Our intuitive notions of the likelihood of events become less reliable when considering the probability of events that are very rare. Are you more likely to get struck by lightning or be bitten by a rabid animal? Both may be life-changing (perhaps life-ending) rare events, but which should you be more concerned about? If you were a public official, would you invest in preventing lightning strikes or rabid animal bites?

To quantify the probability of rare events we use the same methods of physical symmetry or past frequency. However, quite often we have evaluated the probability of an event happening per day or per year, but want to know the chance of it happening over a longer period of time. On a random day your chance of being struck by lightning is approximately 3.5×10^{-9} , but what is the probability of being struck at least once in your lifetime?

Let us perform the calculation of the probability of a more cheerful rare event, the chance of winning the lottery. From physical symmetry considerations of how lottery numbers are chosen, one can calculate exactly that the chances of a \$1 ticket winning the New York Mega Millions jackpot are 1 in 175,711,536 or $p(\text{lottery}) = 5.7 \times 10^{-9}$. Playing once, it seems highly unlikely you will win, but suppose you play the twice-weekly game every week for the next 50 years, a total of around 5,000 attempts. Since each event, i.e. each lottery drawing, is independent, we can use the rule for independent probabilities to calculate the probability of winning at least once:

$$P(\text{winning the lottery in 5,000 attempts}) = 1 - (1 - p(\text{lottery}))(1 - p(\text{lottery}))(1 - p(\text{lottery})) \dots$$

$$= 1 - (1 - p(\text{lottery}))^{5,000}$$

$$= 1 - (1 - 5.7 \times 10^{-9})^{5,000} = 2.8 \times 10^{-5}$$

or ~1 in 35,000 chance.

Subtraction of small numbers and raising numbers to large powers are both cumbersome,[§] but there is a useful approximation in calculating rare events. If instead of using the exact formula we had multiplied the number of attempts with the probability of winning for one attempt we would have computed

[§] Tip: don't round off your answer until you reach the end of the calculation. The first step of the above calculation is $1 - 5.7 \times 10^{-9} = 0.9999999943$. If we had rounded this up to 1, the final answer for the probability would have been $1 - 1^{5000} = 0$. Although 2.8×10^{-5} is very small, it is distinctly different from zero.

$(5.7 \times 10^{-9})(5,000) = 2.9 \times 10^{-5}$, almost the same as the more complicated calculation above. The simpler computation, of multiplying the probability of one attempt by the number of attempts is an excellent approximation for the exact calculation *provided* the product is much less than 1. Summarizing:

If $p(A)$ is the probability of event A occurring given one opportunity, then the probability of event A occurring at least once given N opportunities is given by:

$$P = 1 - (1 - p(A))^N$$

If this probability is very small, it is well approximated by:

$$P = N \cdot p(A)$$

Ignoring the first clause produces erroneous results: there is a 1/6 chance of rolling a 2 for one roll of a six-sided die, but there is not a $10 \times 1/6 = 166\%$ probability of rolling a 2 in 10 rolls, just as there is not a $6 \times 1/6 = 100\%$ probability of rolling a 2 in 6 rolls. For these types of problems the more cumbersome, but exact formula must be used.